TuteDude

ASSIGNMENT 1: SQL MASTERY - The E-Commerce Analytics Challenge

Topics Covered

SQL Manipulation, Retrieval, Aggregation, Joins, Subqueries, CTEs, Window Functions, Stored Procedures, Indexes

Duration: 7–8 Hours
Questions: 30

Problem Statement

You're hired as a Data Engineer at "ShopHub" – a fast-growing e-commerce platform. The company has raw transactional data scattered across multiple systems. Your job is to design, optimize, and create a complete analytics database.

Dataset

Brazilian E-Commerce Public Dataset by Olist

Dataset Link:
https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce

Alternative Source:
https://github.com/olist/brazilian-ecommerce

Dataset Size: ~100K orders, 8 CSV files (~45 MB)

Files Included:

- customers
- orders
- order_items
- products
- sellers
- order_payments
- order_reviews
- geolocation

Assignment Questions

Part A: Database Design & Data Quality (5 Questions)

1. Design a normalized database schema (3NF) with ERD showing all relationships between 8 tables.
2. Identify and document 10+ data quality issues in the raw CSV files.
3. Create tables with appropriate data types, primary keys, foreign keys, and constraints.
4. Write a SQL script to detect and remove duplicate customers while keeping the most recent record.
5. Create a comprehensive data dictionary documenting each table, column, data type, and business meaning.

Part B: Data Manipulation & Retrieval (5 Questions)

6. Insert cleaned data using transactions and demonstrate COMMIT and ROLLBACK.
7. Find customers who have the same email but different names/addresses.
8. Identify orphan records referencing non-existent products or orders.
9. List customers who registered but never placed an order.
10. Detect potential fraud where payment value differs from order total.

Part C: Complex Aggregations (5 Questions)

11. Calculate monthly revenue with Month-over-Month growth percentage.
12. Find top 10 products by revenue in each category.
13. Calculate Customer Lifetime Value (CLV) and categorize customers.
14. Create sales reports using ROLLUP or CUBE.
15. Identify seasonal sales patterns by product category.

Part D: Mastering Joins (5 Questions)

16. Create a customer 360-degree view joining multiple tables.
17. Find customers who bought electronics but never books.
18. List sellers and their best-selling product in each category.
19. Perform market basket analysis using self-joins.
20. Identify orders with shipping delays.

Part E: Subqueries & CTEs (4 Questions)

21. Find customers spending above their state's average.
22. Get the 2nd highest revenue-generating product in each category.
23. Create a product category hierarchy using recursive CTEs.
24. Find customers purchasing in 3+ consecutive months.

Part F: Advanced Window Functions (4 Questions)

25. Calculate a 7-day moving average of daily order count.
26. Find gaps between consecutive customer orders using LAG().
27. Rank sellers by revenue within each state.
28. Calculate running revenue totals and percentage contribution.

Part G: Stored Procedures & Optimization (2 Questions)

29. Create a stored procedure for dynamic discount calculation.
30. Optimize slow queries using EXPLAIN ANALYZE and indexes.

Submission Guidelines

1. Prepare Assignment Folder

Create a folder named:

SQL_Mastery_Assignment_1

Add the following files:

Required Files

- ERD Diagram (PNG, JPG, PDF, or Draw.io export)
- SQL Scripts
    - 01_schema
    - 02_data_cleaning
    - 03_queries
    - 04_optimization
- Data Quality Findings Report (PDF or Markdown)
- Performance Optimization Report with EXPLAIN outputs
- Screenshot Evidence of completed work (queries executed, outputs, ERD, reports, etc.)
- README file (optional but recommended)

2. Create ZIP File

After organizing all files:

- Compress the assignment folder into a ZIP file.
- Example:
    - SQL_Mastery_Assignment_1.zip

3. Verification Checklist

Before submission, ensure:

- All SQL scripts are included.
- ERD diagram is attached.
- Reports are complete and readable.
- Screenshots clearly show query execution and outputs.
- ZIP file opens correctly without errors.

4. Submission

Upload the ZIP file in the designated assignment submission section.

Note

You can always connect with your mentor through the 1:1 Live Mentorship support option if you have any doubts or queries regarding the assignment.